Fuzzy Recovery: Fast and Resilient Approach to Post-Attack Restoration for Transactional Systems

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Abstract

Critical infrastructure systems require rapid post-attack recovery, but traditional mechanisms prioritizing perfect accuracy are too slow. This paper presents a fuzzy recovery framework that balances speed and correctness using statistical range estimation and real-time anomaly detection. We define three value sets—Accepted, Possibly Correct, and Accurate—enabling flexible recovery strategies based on urgency. Triangular fuzzy membership functions score transaction plausibility, maintaining operational continuity while preserving data integrity. The proposed fuzzy recovery system uses dual-processor architecture which separates normal processing from parallel recalculation. When anomalies are detected (membership score 0.5), a secondary processor computes provisional values using historical means while an Intrusion Detection System verifies threats, avoiding system-wide halts. Upon confirmation, pre-computed corrections are applied, followed by log-based recovery for perfect accuracy. Simulations across 100 accounts over 30 days show the fuzzy system maintained substantial throughput during recovery while traditional systems imposed complete halts. The results demonstrate that fuzzy recovery achieves an effective balance between responsiveness and correctness, offering a practical and resilient alternative for high-volume transactional environments.

CCS Concepts

• **Theory of computation → Data structures and algorithms for data management**.

Keywords

Database Damage Assessment, Recovery, Membership Function

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1 Introduction

The importance of efficient damage assessment and recovery in database systems cannot be overstated, particularly in critical environments such as financial institutions, healthcare facilities, and air traffic control systems. In these scenarios, quickly identifying and isolating damaged data is paramount, as prolonged downtime or full system shutdowns are often not viable options. Traditional methods of addressing data integrity issues frequently involve scanning entire log files—an approach that becomes increasingly time-intensive and resource-demanding as databases scale and complexity escalates [7, 10, 17].

In post-attack or failure scenarios, the rapid restoration of system functionality is crucial, particularly when dealing with critical infrastructure. Traditional recovery mechanisms that focus on perfect accuracy often prove too slow to meet the urgent demands of such systems. Critical infrastructure systems, such as financial networks and power grids, are highly interconnected and rely heavily on timely, accurate data. These systems typically store sequential logs on slow secondary storage, making comprehensive post-attack recovery a lengthy process. When rapid recovery and restoration of normalcy is a top priority, such inherent slowness is not acceptable. In urgent situations, waiting for a full reconstruction of accurate values is often impractical, raising the question of how to restore critical operations with data that are safe to use, even if they are not perfectly accurate.

To address this challenge, a hybrid recovery framework can be employed, leveraging fuzzy logic and statistical range estimation to balance the need for both speed and correctness. This framework is built upon two primary types of recovery processes: perfect recovery and fuzzy recovery. Perfect recovery aims to reconstruct the exact value that would have existed had there been no attack, thereby restoring the system to a state of perfect integrity with no possibility of error. However, this approach is slow, as it requires reading and executing all operations from the log in their original sequence, which is often impractical in urgent scenarios. In contrast, fuzzy recovery is designed to provide values that may not be exact but are still operationally acceptable. These values enable the system to quickly resume critical functions, with the understanding that more precise corrections can be made later if necessary.

The proposed approach establishes three classes of value sets: Accepted Values, Possibly Correct Values, and Accurate Values. By defining these categories, a spectrum of recovery strategies is created, allowing flexibility depending on the severity of the situation and the time constraints involved. Fuzzy scoring techniques are used within this framework to detect anomalies in data and

to suggest the most appropriate recovery path, ensuring that the system can resume operations quickly and safely, even when some uncertainty remains.

Paper Outline: Section 2 reviews the current literature on database security, intrusion recovery systems, and fuzzy logic applications in cybersecurity. Section 3 describes the methodology, detailing the fuzzy recovery framework, triangular membership functions, and the dual-processor architecture for real-time anomaly detection and recovery. Section 4 presents the simulation results with synthetic transaction datasets to demonstrate system throughput and resilience under adversarial scenarios. Finally, Section 5 concludes the paper.

2 Related Work

Database systems serve as repositories for organizations' most critical and high-value information assets. The proliferation of database usage has been accompanied by a corresponding increase in targeted attacks that threaten data integrity and availability [5, 11]. These attacks seek to access, manipulate, or destroy sensitive information including financial records, authentication credentials, and personal identifiers. Exploiting architectural vulnerabilities [1, 5, 12] in database systems, attackers employ diverse techniques ranging from SQL injection to privilege escalation. When such attacks succeed, rapid damage assessment and recovery become a huge necessity—requiring systems that can not only detect malicious transactions but also classify their severity and trace their propagation through transaction dependencies to minimize system downtime and data loss

Database security has become increasingly critical as organizations rely on databases to store sensitive corporate and personal information. Recognizing that the value of data necessitates robust protection mechanisms, [14] undertook a systematic mapping study (SMS) to identify and highlight the difficulties businesses face in establishing, implementing, and maintaining secure local and worldwide databases. The study examined security issues across the software development lifecycle (design, development, and maintenance), types of database attacks, and publication venues, capturing current research trends and diverse challenges from a software engineering perspective.

Building on these security challenges, the forensic analysis of database artifacts has emerged as a critical investigative tool. As a specialized field of digital forensics, database forensics focuses on investigating database-related incidents, including data tampering, breaches, and leakages [9]. From a forensic perspective, the authors of [8] conducted groundbreaking research on Microsoft SQL Server transaction log analysis, presenting methods to identify and extract transaction log records from unallocated areas of file systems. Their work addresses a significant gap in database forensics, as previous raw-level studies focused primarily on Oracle and MySQL transaction logs, with limited research on MSSQL.

The inherent uncertainty in distinguishing malicious activities from legitimate operations has motivated researchers to explore fuzzy logic as a detection mechanism. Kupchyn et al. [16] proposed a novel validation approach for fuzzy logic prognostic models when experimental data is unavailable, using neural network comparison as a baseline. Their method employs seven hierarchical Fuzzy Inference Systems (FIS) with Mamdani algorithm, processing fourteen input indicators through various membership functions (singleton, Gaussian, and Z/S-shaped) to perform classification tasks.

Extending fuzzy logic to cybersecurity risk assessment, Alali et al. [6]proposed a Mamdani-based Fuzzy Inference System (FIS) for cybersecurity risk assessment to address the challenges of evaluating security threats in environments with vague and uncertain information. The model employs a hierarchical architecture with three cascaded FIS modules that process four primary risk factors: vulnerability, threat, likelihood, and impact. The first FIS (F1) evaluates overall attacker capabilities by combining intent, targeting, and capabilities using triangular and trapezoidal membership functions with five linguistic categories (Very Low, Low, Medium, High, Very High). The output from F1 feeds into the second FIS (F2), which combines overall capabilities with system vulnerabilities to determine overall likelihood, while the third FIS (F3) integrates overall likelihood with impact to produce the final risk assessment value. The Mamdani approach was selected over Sugeno due to its intuitive linguistic output, human-friendly interpretation, and suitability for qualitative security analysis, despite Sugeno's efficiency in optimization scenarios.

Demonstrating practical application in intrusion detection, [21] developed a simulation model (FIDM - Fuzzy Intrusion Detection Model) to detect cyberattacks, including new variations that the system has never seen before. The model analyzes 38 different network traffic characteristics (such as connection time, number of bytes transferred, and login attempts) to identify five types of threats: Denial of Service attacks, Remote to Local intrusions, User to Root escalations, Probe attacks, and Normal system behavior. The system works in two stages: first, it uses traditional signature matching to catch known attacks, and second, it applies fuzzy logic to detect suspicious patterns that might indicate new or modified attacks. Triangular membership functions play a critical role in this approach because they allow the system to represent expert knowledge about what constitutes "normal" versus "suspicious" behavior while being computationally efficient and easy to adjust through automated learning.

Building on fuzzy logic principles for intrusion detection, [13] explored the application of fuzzy logic in intrusion detection systems (IDS), demonstrating its effectiveness in addressing uncertainty in network traffic patterns. Their research utilized the NSL-KDD dataset and compared fuzzy logic models employing triangle and parallelogram-shaped membership functions against conventional machine learning models such as Decision Trees and Support Vector Machines. The fuzzy logic system achieved superior performance with greater accuracy, precision, and F1-scores than conventional models, particularly when applied to real network traffic data. The authors also incorporated Wasserstein Generative Adversarial Networks (WGANs) to generate synthetic data for privacy-preserving evaluation, demonstrating the model's robustness and feasibility for intrusion detection in sensitive contexts.

When intrusions are detected, efficient recovery depends on well-structured logging mechanisms. ClusteredLog is introduced in [4], a novel approach to optimizing log structures for efficient data recovery and integrity management in database systems. Traditional

log structures store operations sequentially as they occur, leading to significant delays in accessing and recovering specific data objects due to their scattered nature across the log. ClusteredLog addresses these limitations by implementing a logical organization that groups related operations for each data item into clusters, thereby enabling faster identification and recovery of damaged data items.

Beyond optimized logging, accurate damage assessment is essential for targeted recovery. A comprehensive framework for damage assessment in fog computing systems using a blind write lineage approach is devoloped in [2] and simulated in [3]. Fog computing systems are particularly vulnerable due to their large number of interconnected nodes, where damage from attacks can rapidly spread to other nodes when valid transactions use compromised data values. The research addresses critical challenges in real-time processing environments, where traditional recovery methods involving system shutdowns are not viable.

For time-critical environments requiring minimal downtime, parallelized damage assessment approaches offer significant advantages. A parallelized database damage assessment approach for healthcare systems is introduced in [15], where minimal downtime is critical to patient safety. Inspired by COVID-19 infection behavior, the authors proposed a multi-threaded algorithm that treats malicious transactions as viruses and affected transactions as infected individuals requiring isolation and testing. The algorithm employs a transaction dependency paradigm where threads execute in parallel across different database areas, each checking whether transactions in its designated region have interacted with malicious or affected data items. The method maintains a synchronized testing list that is continuously updated as threads identify newly affected transactions, terminating when all threads reach the end of the list with no new infections detected.

These damage assessment techniques inform broader intrusion recovery strategies. A systematic survey of intrusion recovery systems spanning 25 years is conducted in [18], addressing the critical limitation that traditional backups permanently lose legitimate changes made since the last backup. The authors taxonomized recovery approaches into five domains (generic applications, virtual machines, multiversioned file systems, selective re-execution systems, and web/cloud applications) and distinguished between backward recovery (reverting entire systems) and forward recovery (selectively undoing only malicious operations).

However, traditional recovery approaches assume the integrity of audit and transaction logs. When these logs are compromised, alternative forensic methods become necessary. MemTraceDB [19], a memory forensics tool that reconstructs user activity timelines when traditional database audit and transaction logs have been compromised by privileged attackers. Recognizing that malicious insiders can alter, purge, or temporarily disable logging mechanisms, the author developed a methodology that analyzes volatile memory artifacts as a ground-truth source of database activity, bypassing unreliable disk-based logs. MemTraceDB employs the ActiviTimeTrace algorithm to extract forensic artifacts from MySQL process memory snapshots, including user connections, executed queries, and query execution sequences through four data structures: user connection information, user system information, recent queries per user, and a global query stack. Through experimental validation, Nissan discovered that the MySQL query stack has a finite capacity of approximately 9,997 queries, after which the system overwrites the oldest query, enabling establishment of an optimal memory snapshot collection frequency formula.

For web application environments with complex database and file system interactions, specialized recovery mechanisms address unique attribution challenges. ANCORA [20], a system for precise intrusion recovery in web applications that addresses the critical challenge of attributing database and file operations to specific malicious requests in high-concurrency environments. Recognizing that coarse-grained snapshot rollbacks cause unacceptable data loss for legitimate users, the authors developed a surgical recovery approach using syscall-based provenance graphs and a novel spatiotemporal anchor technique for database operation attribution. ANCORA employs a three-stage workflow: request partition to isolate individual request syscalls, post-attack file system recovery using execution dependency analysis to construct causal chains revealing all file modifications, and post-attack database state recovery leveraging network connection tuples and precise time windows to pinpoint exact database operations.

3 Methodology

The fuzzy recovery model addresses this by defining three sets of values. Accepted Values are those that are statistically plausible and can be computed rapidly, enabling immediate operational continuity. Possibly Correct Values are derived by analyzing the plausible effects of an attack, providing a more refined but computationally intensive recovery option. Accurate Values represent the ideal state, obtained through exhaustive analysis, but are much slower to determine.

Definition 1 (*Accepted Value Set* ($\mathcal{A}$)): The Accepted Value Set, $\mathcal{A}$, contains values considered operationally safe. These are derived quickly through statistical profiling of historical values:

If $x_{\min}$ and $x_{\max}$ represent the observed historical minimum and maximum of field x, then:

$$\mathcal{A} = \{x \mid x_{\min} \leq x \leq x_{\max}\}$$

These values may not be accurate, but they allow the system to continue operating safely while more accurate recovery is pending.

Definition 2 (*Possibly Correct Value Set* $\mathcal{P}$): The Possibly Correct Value Set, $\mathcal{P}$, is a refined subset of $\mathcal{A}$, statistically centered around the accurate value. It is defined using the historical mean μ and standard deviation σ:

$$\mathcal{P} = \{x \mid \mu - k\sigma \leq x \leq \mu + k\sigma\}$$

Here, k is a tunable confidence parameter (typically $k = 1$ or $k = 2$) that controls the precision.

While $\mathcal{P}$ requires more computational resources than $\mathcal{A}$, it offers greater accuracy and is valuable in systems where moderate delays are acceptable.

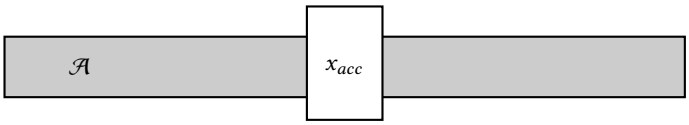
Definition 3 (*Accurate Value* x_{acc}): The Accurate Value, denoted x_{acc}, is the true, uncorrupted value that would exist if no attack or data corruption had taken place. It is computed by applying a full recovery function f over the verified, complete transaction log:

$$x_{\text{acc}} = f(\text{Full Log})$$

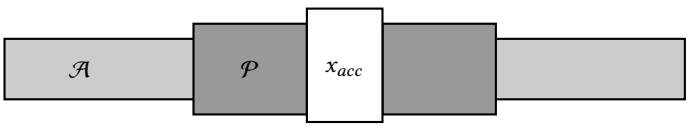
This value is the gold standard in recovery, offering perfect precision. However, it is often computationally expensive and time-consuming, especially in high-volume or compromised environments. Also,

$$x_{\text{acc}} \in \mathcal{P}$$

Where, x_{acc} — the true, correct, uncorrupted value of the data — lies within the Possibly Correct Value Set $\mathcal{P}(x)$.



(a) Fuzzy Recovery with Acceptable Value



(b) Fuzzy Recovery with Possible Correct Values

Figure 1: Fuzzy Recovery Approaches

Figure 1a and Figure 1b illustrate this refinement. Here, the set of accepted values is shown as a broad range of light gray color, while the set of possibly correct values forms a narrower band around the accurate value in dark gray color. The fuzzy recovery process, in this case, returns a value within the possibly correct set, offering a better approximation to the true value which is x_{acc}, while maintaining efficiency. This approach is particularly advantageous when the system requires a higher degree of precision but cannot afford the time required for perfect recovery.

The need to balance accuracy and recovery speed becomes especially clear when examining transaction logs after an attack. For example, if a withdrawal of 2000 units appears in an account where historical withdrawals never exceeded 500, this is a clear anomaly. However, determining the exact pre-attack value through perfect recovery could take too long to be practical. By defining a "possible correct range" based on domain-specific modeling of potential damage, it becomes possible to make faster and safer approximations. In safety-critical environments like healthcare or energy systems, quickly returning a value that is "good enough" may be preferable to waiting for perfect data, since maintaining operational continuity can outweigh the risks associated with minor inaccuracies. This practical consideration is at the core of the fuzzy approach, which is designed to uphold system integrity and service availability even under adverse conditions.

Definition 4 (*Fuzzy Recovery and Membership Function*): The Fuzzy Recovery approach selects a value x_{rec} from $\mathcal{P}$, depending on system constraints:

$$x_{\text{rec}} \in \mathcal{P}(x)$$

This decision is guided by a fuzzy membership function $\mu(x)$, which assigns a trust score to each candidate value x, representing its degree of plausibility.

3.1 Triangular Membership Example

A simple triangular membership function over $\mathcal{A}$ is:

$$mf(x) = \begin{cases} 0 & x < x_{\min} \text{ or } x > x_{\max} \\ \frac{x - x_{\min}}{\mu - x_{\min}} & x_{\min} \leq x \leq \mu \\ \frac{x_{\max} - x}{x_{\max} - \mu} & \mu < x \leq x_{\max} \end{cases}$$

This function peaks at the mean μ and tapers off toward the bounds of $\mathcal{A}$, assigning:

- $mf(x) = 1$ for the most plausible values
- $0 < mf(x) < 1$ for partially trusted values
- $mf(x) = 0$ for values outside the accepted domain

Suppose the historical withdrawal amounts in a transaction log are between 10 and 500 units ($x_{\min} = 10$, $x_{\max} = 500$), with a mean of 250 ($\mu = 250$) and a standard deviation of 100 ($\sigma = 100$). The possibly correct set for $k = 1$ is $150 \leq x \leq 350$.

- If a withdrawal of 2000 appears, it falls outside both the accepted range ($10 \leq x \leq 500$) and the possibly correct range ($150 \leq x \leq 350$), so its membership score is $mf(2000) = 0$. This value should be flagged as anomalous.

In such cases, to ensure the continuity of system operations while a perfect recovery is being performed, we can offer the average (mean) $\mu = 250$, as a temporary substitute. This approach exemplifies fuzzy recovery, where a statistically plausible and operationally safe value is provided in place of the true value. By temporarily substituting the mean, the system can continue to function effectively and securely, minimizing disruption. Once the full log analysis is complete and the accurate value (x_{acc}) is determined, the temporary value can be replaced to restore perfect correctness. This strategy allows for a practical trade-off between immediate operational needs and eventual data integrity, demonstrating the flexibility and utility of fuzzy recovery in real-world scenarios.

- In contrast, a withdrawal of 300 happens, which falls within both $\mathcal{A}(x)$ and $\mathcal{P}(x)$, and its membership score $mf(300) = 0.8$. This value could be used for fuzzy recovery, allowing the system to resume operations quickly and safely while checking the log if this transaction is valid or not.

This framework enables two main recovery models. Perfect recovery reconstructs x_{acc} by analyzing the full log, ensuring maximum accuracy but requiring significant time than fuzzy recovery. Fuzzy recovery selects μ as x_{rec}, prioritizing speed while maintaining operational safety. This allows the system to adaptively trade off between speed and accuracy, ensuring that critical operations can continue even when perfect data is temporarily unavailable. Now let's see how this theory can be used with a dual processor model for fuzzy recovery.

3.2 Real-Time Fuzzy Recovery Strategy with Dual-Processor Model

To enhance the fuzzy recovery model for critical systems, we propose a real-time transaction monitoring system supported by a dual-processor setup that performs fuzzy validation and recalculation in parallel. This approach allows immediate response to suspicious transactions without halting system operations.

3.2.1 Real-Time Monitoring with Triangular Membership Function. During each transaction, as the system processes numerical input values (e.g., financial amounts, sensor readings, etc.), it continuously evaluates the membership score of each transaction using the triangular membership function (denoted as $mf(x)$).

3.2.2 Real-Time Fuzzy Validation and Recovery Logic. This section formalizes the real-time validation and recovery process by detailing how each incoming numerical transaction value is assessed using a dual-processor architecture and fuzzy membership scoring.

Fuzzy system will keep monitoring the transactions continuously. For every new transaction T_i, the system immediately computes its membership score, denoted as $mf(x)$, using the triangular membership function. This score reflects how "normal" or "plausible" the value is based on historical patterns.

If a data item's updated value falls completely outside the accepted historical range, it is considered anomalous or suspicious. This modified value which was generated during a transaction will raise immediate concern as it significantly deviates from the known behavior of that data item, triggering two actions in parallel:

- Parallel Recalculation (Secondary Processor) : A secondary processor computes a temporary balance by replacing the suspicious value with the historical mean, ensuring the system can continue functioning without delay.
- Intrusion Detection System (IDS) Alert : Upon detection, the IDS is immediately notified to investigate the flagged transaction. If malicious activity is confirmed, the mean-substituted result remains in use as the operational value to maintain system continuity. Concurrently, a recovery process scans the transaction log to locate the accurate value. Once recovered, this accurate value is inserted into the transaction log and propagated forward to recompute all values that were affected by the temporary substitution.

If a transaction's score is low but not zero (Borderline Suspicion, $mf(x) \leq 0.5$), the IDS is not overwhelmed but processes the alert asynchronously. The secondary processor still performs a parallel recalculation as a precaution. The final decision to replace the value is made only if the IDS confirms a potential threat.

For transactions with high membership scores that is for trusted transactions ($mf(x) > 0.5$), the system proceeds normally without interruption, avoiding any recalculation or IDS alerts.

This design decision is justified because such values are within or near the Possibly Correct Set, closely matching the expected behavior and are rarely malicious. If later proven malicious, the system recovers the correct value from logs, but at a cost of delayed correction. This reduces IDS workload and avoids unnecessary computations.

3.3 Case Study: Fuzzy Recovery Workflow in Practice

To better demonstrate the effectiveness of our fuzzy recovery strategy, let us consider a real-world transaction scenario involving a customer account over several days (Table 1). The historical statistics (see Table 2) show the accepted and possible correct value ranges for each data field—Deposit and Withdrawal—derived from minimum, maximum, mean, and standard deviation (STD) values.

The Accepted Value Range reflects the historical minimum and maximum values observed for each transaction type (Deposit and Withdrawal) using the annotation of [Min, Max]. Any value within this range is treated as statistically acceptable.

The Possible Correct Value Range is derived using the average (Avg) ± standard deviation (STD) for each field. This defines a more refined boundary that captures the "possibly correct set".

For instance (Table 2), the accepted range for Deposits is [68.2, 486.46], while the possible correct range is derived as [144.2646, 386.3054] using the formula [Avg − STD, Avg + STD]. Any new transaction falling outside the accepted range ($mf(x) = 0$) or even on the borderline of the possible correct range ($0 < mf(x) \leq 0.5$) raises suspicion.

In the transaction log presented in Table 1, let's look at the customer 12345 on June 27, 2025. A Withdrawal of 2000 is detected, which clearly falls outside the accepted range for withdrawals ([10.09, 496.26]). Using the triangular membership function, this value yields $mf(2000) = 0$, marking it as an outlier. As a result, the system triggers the fuzzy recovery process in real time:

- IDS is notified immediately, and in parallel, a secondary processor begins recalculating the current balance using the historical mean value of withdrawal (237.00) in place of the suspicious amount.
- This recalculated balance is used temporarily in the primary processor once IDS verified it is malicious to allow all subsequent transactions to continue without interruption. As shown in Table 1, the secondary processor generates updated balances using the mean-substituted value across several days.

In this scenario, upon confirmation of a malicious transaction by the IDS—such as the one on June 30, 2025, when the customer's balance reaches 411.22—a notification is immediately sent to the customer, informing them of the detected threat and the suspension of all account activities pending resolution. The system then replaces the live current balance with the precomputed, recalculated value (2174.22) sourced from the secondary processor. This updated baseline becomes the new reference point for the customer's account, allowing safe continuation of subsequent transaction processing following recovery.

Upon identification and removal of the malicious transaction—specifically the withdrawal of 2000 on 06/27—the system retrieves the correct historical record from the transaction log and initiates recalculation of all subsequently affected balances (as illustrated in Table 3). The erroneous entry is replaced, and each impacted balance is updated, propagating the corrected values forward throughout the customer's transaction history. This staged process, first isolating the suspicious transaction and then restoring full consistency through targeted recalculation, guarantees real-time operational continuity alongside long-term accuracy and data integrity.

This example illustrates the model's operational advantages. The system performs real-time substitution of the suspicious value with a statistically safe placeholder (mean) to avoid delays, maintains minimal disruption by proceeding without waiting for IDS validation, and ensures precise correction post-validation when the malicious transaction is identified and deleted from logs.

Table 1: Daily Transaction Log Snapshot for a Sample Customer Account for a Month

Primary Processor						Secondary Processor					
CustID	**Date**	**PrevBal**	**Dep**	**With**	**CurrBal**	Idle as any $mf(x) > 0.5$ of deposit or withdrawal was					
12345	06/01	1000	N/A	476.34	523.66						
12345	06/01	523.66	400.86	N/A	924.52						
			⋮								
12345	06/26	2023.57	242.39	N/A	2265.96	**CustID**	**Date**	**PrevBal**	**Dep**	**With**	**CurrBal**
12345	06/27	2265.96	N/A	*2000*	265.96	12345	06/27	2265.96	N/A	*237*	2028.96
12345	06/27	265.96	453.09	N/A	719.05	12345	06/27	2028.96	453.09	N/A	2482.05
12345	06/28	719.05	N/A	290.37	428.68	12345	06/28	2482.05	N/A	290.37	2191.68
12345	06/29	428.68	162.18	N/A	590.86	12345	06/29	2191.68	162.18	N/A	2353.86
12345	06/29	590.86	292.93	N/A	883.79	12345	06/29	2353.86	292.93	N/A	2646.79
12345	06/30	883.79	N/A	472.57	*411.22*	12345	06/30	2646.79	N/A	472.57	*2174.22*

Table 2: Accepted and Possible Correct Value Ranges for Deposit and Withdrawal

Accepted Value Range		Possible Correct Value Range	
	Values		**Values**
Deposit	[68.2, 486.46]	**Deposit**	[144.2646, 386.3054]
Withdrawal	[10.09, 496.26]	**Withdrawal**	[73.389, 400.6126]

Table 3: Reconstructed Transaction Log After Fuzzy Recovery

CustID	Date	PrevBal	Dep	With	CurrBal
12345	06/01	1000	N/A	476.34	523.66
12345	06/01	523.66	400.86	N/A	924.52
			...		
			...		
12345	06/26	2023.57	242.39	N/A	2265.96
~~12345~~	~~06/27~~	~~2265.96~~	~~N/A~~	~~***2000***~~	~~***265.96***~~
12345	06/27	***2265.96***	453.09	N/A	***2719.05***
12345	06/28	***2719.05***	N/A	290.37	***2428.68***
12345	06/29	***2428.68***	162.18	N/A	***2590.86***
12345	06/29	***2590.86***	292.93	N/A	***2883.79***
12345	06/30	***2883.79***	N/A	472.57	***2411.22***

Unlike traditional systems that allow suspicious transactions—such as the withdrawal of 2000—to proceed unchecked and later require a costly rollback of all dependent operations, our model preemptively isolates the anomaly, performs a parallel recalculation, and prepares for both benign and malicious outcomes, thereby minimizing disruption and recovery overhead.

3.4 Comparison with Traditional Systems

Traditional systems normally do not monitor deviations in real-time. As a result, all transactions, including potentially malicious ones, are processed. When an attack is discovered, the system must retroactively recover all dependent transactions, which is time-consuming. Our fuzzy model addresses this by dividing transactions into three distinct cases based on $mf(x)$:

Case 1 ($mf(x) > 0.5$) : Transactions are treated the same as traditional systems. Recovery is only required if an extremely rare malicious transaction is later identified by the IDS. In such scenarios, the recovery approach aligns with that of traditional systems.

Case 2 ($0 < mf(x) \leq 0.5$) : To prevent overwhelming IDS, it is not immediately alerted in this case; instead, the system continues processing while waiting for the IDS to verify the transaction as the secondary processor recalculates in the background. If the IDS confirms it is malicious, the precomputed value is used immediately, saving time. If not malicious, the recalculated value is discarded with no further action required.

Case 3 ($mf(x) = 0$) : The IDS is immediately alerted for urgent verification, and the secondary processor simultaneously performs recalculation using the historical mean. Once confirmed as malicious, recovery is achieved faster than both traditional systems and Case 2, since IDS starts analysis immediately and the system already has the corrected value available for immediate use.

This intelligent division and proactive recalculation approach makes the system both efficient and resilient, ensuring minimal disruption while maintaining long-term data accuracy.

3.5 Why Dual Processors? A Time vs Space Tradeoff

The system employs two processors: the primary processor handles normal transactions and system continuity, while the secondary processor handles recalculations and fuzzy substitutions.

Benefits: The dual-processor architecture delivers three key advantages: accelerated response by bypassing full log recovery, minimized downtime during recovery operations, and enhanced system resilience under attack or anomalous conditions.

This setup uses more hardware (space) but saves significant time, allowing critical systems to continue operating with minimal delay.

Assumptions: The model operates under two key assumptions. First, attackers cannot access system logs or reverse-engineer the fuzzy inference system. Second, attackers are not regular users and therefore lack insider knowledge of transaction patterns. While a long-term user might estimate minimum, maximum, or average values, such familiarity is unlikely in most attack scenarios.

This real-time fuzzy strategy ensures safety, continuity, and resilience, even when some uncertainty is present. It minimizes disruption, maximizes responsiveness, and prepares the system for eventual perfect recovery without sacrificing short-term functionality.

3.6 Transaction Model and Data Structure

In our model, each transaction T_i is not limited to a single value change. Instead, it can involve a sequence of operations affecting multiple data fields—such as deposits, withdrawals, balances, and timestamps in banking or financial contexts. We define the set $\mathcal{X} = \{x_1, x_2, x_3, \ldots, x_j\}$ to represent all such data items that may be accessed or modified by T_i. Each data item $x_j \in \mathcal{X}$ can be updated multiple times within a single transaction.

$$T_i = \{x_1, x_2, x_3, \ldots, x_n \mid x_j \in \mathcal{X}\}$$

This structure is critical for fuzzy recovery, as the system must evaluate the membership score $mf(x_j)$ for each updated value in real time. If a value is suspicious or anomalous, recovery mechanisms such as recalculation or IDS verification are triggered. The following flowcharts (Figure 2 and Figure 3) outline the step-by-step logic the system follows to validate, recover, and ensure integrity for each transaction during execution. First, let us walk through the process of real-time fuzzy validation.

Figure 2 illustrates the real-time behavior of the system when a transaction T_i arrives and modifies one or more data items x_j from the set $\mathcal{X}$. Each value x_j is evaluated using a triangular membership function $mf(x_j)$. If the membership score suggests abnormality, specifically when $0 < mf(x_j) \leq 0.5$ or $mf(x_j) = 0$, a secondary processor instantly recalculates a plausible substitute (e.g., the historical mean) to maintain uninterrupted transaction flow after IDS verification in Figure 3. Meanwhile, depending on the severity of the membership score:

- If $mf(x_j) = 0$, the system immediately notifies the IDS, treating the value as highly suspicious. The IDS evaluates the value to determine whether it is malicious, while the system simultaneously resumes normal transaction processing.
- If $0 < mf(x_j) \leq 0.5$, the transaction is considered borderline suspicious. In this case, the system relies on the IDS to verify in the background whether the transaction is malicious, allowing normal operations to continue without immediate disruption.
- If $mf(x_j) > 0.5$, the transaction is deemed trustworthy and is processed normally unless later flagged by the IDS.

Regardless of the outcome, recalculated values are temporarily held in the secondary processor. Depending on the IDS outcome, either the primary processor continues using these provisional values or they are discarded. The detailed follow-up actions, including correction and accurate value restoration from the log, are addressed in Figure 3.

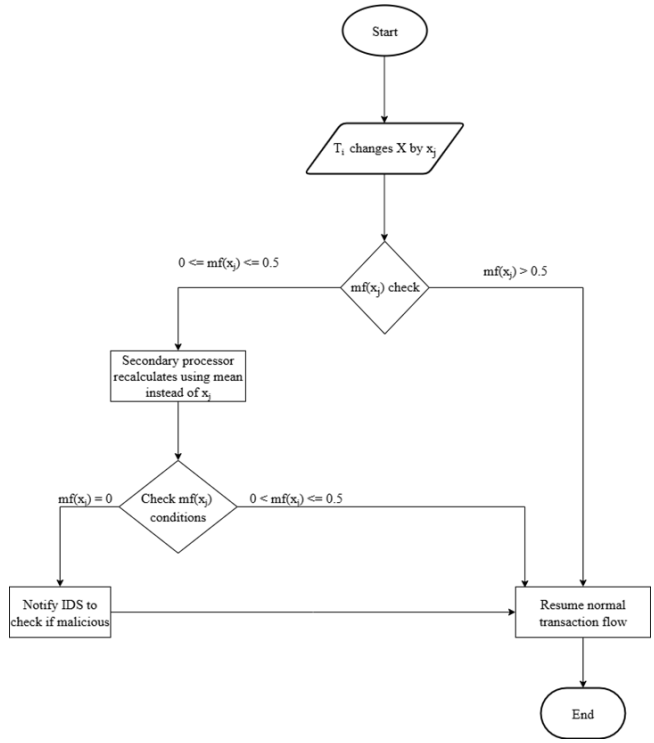


Figure 2: Real-Time Fuzzy validation

Next, we examine the flowchart outlining the IDS verification process and the corresponding backend correction and recovery mechanism.

Figure 3 illustrates the delayed correction phase, which is triggered after the IDS confirms a transaction T_i as malicious. While the IDS verification is still pending, the primary processor continues executing all transactions as usual to avoid system disruption. Once the IDS confirms the threat, the system initiates a backend recovery process. For cases where $mf(x_j) > 0.5$, the transaction will proceed normally, and no other action is needed. For transactions where $0 \leq mf(x_j) \leq 0.5$, the primary processor had been using a recalculated value, typically the historical mean, to replace the suspicious one and ensure uninterrupted operation.

After the malicious transaction is deleted, the system initiates a comprehensive recalculation of all affected transactions to reflect the restored data integrity. Temporary placeholders that were used during the recovery process are then systematically updated with the accurately recalculated value, ensuring that every balance is returned to its correct state and the overall consistency of the database is maintained. This flow represents the post-real-time correction logic, complementing the immediate handling strategy outlined in Figure 2. Together, these mechanisms maintain both operational continuity and long-term data integrity.

The algorithm for the fuzzy recovery will now be explored.

Algorithm 1 Real-Time Fuzzy Recovery with Dual-Processor Support

Data Structure:

- i-th transaction (T_i): A specific transaction at the i-th position
- $\mathcal{X}$: A generic incoming transaction value (deposit, withdrawal, etc.)
- x_j: The j-th data item or field affected by T_i; a transaction can affect multiple such values $x_j \in \mathcal{X}$
- $x_{j_{\min}}, x_{j_{\max}}$: Historical minimum and maximum for the field
- y_{x_j}: Final value after each T_i transactions using x_j
- μ: Historical mean
- *IDS*: Intrusion Detection System
- *Log*: Full transaction log
- x_{acc}: The historical accurate value from the *Log* to replace a suspicious or incorrect x_j
- T_{mal_id}: Transaction flagged as malicious by *IDS*
- $T_{current_tx}$: The current transaction going on in the primary processor when system gets to know about T_{mal_id}

Initialize:
PrimaryProcessor ← handles normal transactions
SecondaryProcessor ← handles background recalculations
Initialize $x_{j_{\min}}, x_{j_{\max}}, \mu$ for each transaction field
IDS ← Active

for each transaction T_i with value x_j **do**
  Compute $mf(x_j)$ using triangular membership function
  if $x_j < x_{j_{\min}}$ OR $x_j > x_{j_{\max}}$ **then**
    $mf(x_j) \leftarrow 0$
  else if $x_{j_{\min}} \leq x_j \leq \mu$ **then**
    $mf(x_j) \leftarrow \frac{x_j - x_{j_{\min}}}{\mu - x_{j_{\min}}}$
  else if $\mu < x_j \leq x_{j_{\max}}$ **then**
    $mf(x_j) \leftarrow \frac{x_{j_{\max}} - x_j}{x_{j_{\max}} - \mu}$
  end if

  # Case 1: Trusted Value
  if $mf(x) > 0.5$ **then**
    PrimaryProcessor.process(T_i)
    continue
  else if $0 < mf(x) \leq 0.5$ **then**
    # Case 2: Borderline Suspicion
    Flag T_i as borderline suspicious
    SecondaryProcessor.recalculate_balance()
    Wait for IDS (low-priority)
    if IDS confirms malicious **then**
      $T_i \leftarrow T_{mal_id}$
      Replace $y_{x_j} \leftarrow y_{\mu_j}$
      After deleting T_{mal_id}, recalculate all affected T_i
    else
      PrimaryProcessor.process(T_i)
    end if
  else if $mf(x_j) = 0$ **then**
    # Case 3: Highly Suspicious
    Flag T_i as highly suspicious
    Notify IDS (high-priority)
    SecondaryProcessor.recalculate_balance()
  end if
end for

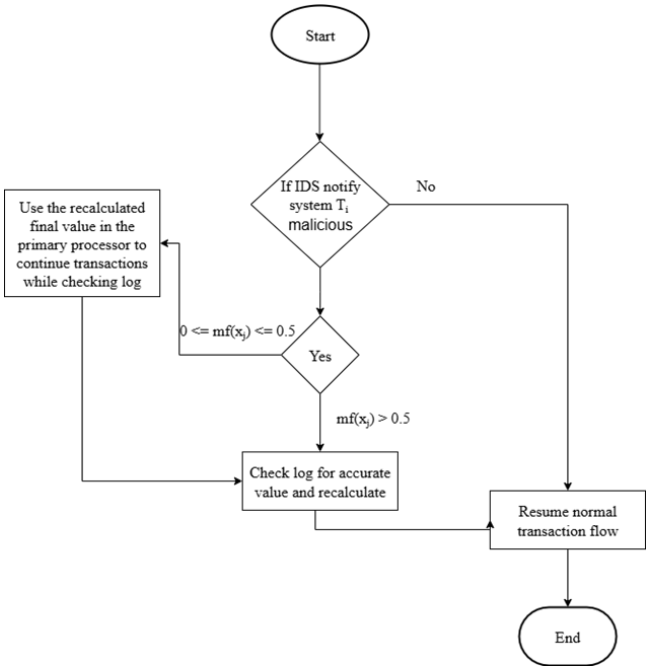


Figure 3: Real-Time Fuzzy validation

4 Simulation Results

4.1 Experimental Setup

To empirically validate the proposed fuzzy recovery framework, we simulated sequential transaction flows across multiple customer accounts under adversarial scenarios. Each transaction was dynamically assessed for plausibility using a triangular fuzzy membership function, whose parameters were continuously tuned on rolling transaction statistics (i.e., min, max, and mean recalculated per customer as new transactions occurred). Transactions falling outside these adaptive statistical boundaries ($mf(x) = 0$ for "Highly Suspicious", $0 < mf(x) \leq 0.5$ for "Borderline") triggered immediate or background IDS alerts (for Highly Suspicious transactions only) and instigated the parallel fuzzy recovery procedures (for both) described previously.

Our fuzzy recovery approach was quantitatively assessed by building a simulation model that replicated activity in a typical transaction-driven infrastructure bank system.

Simulation Parameters

- **Number of customers:** 100 synthetic customers, each assigned a unique CustomerID
- **Duration:** 30 consecutive days, to capture both short-term and longer-term operational behavior and fluctuations
- **Transaction scheduling:** For each customer on each day, the number of transactions is randomly selected between 5 and 10 (drawn from a uniform distribution)

In real critical systems (banking, utilities, etc.), transaction frequency per user is inherently variable. Some accounts may be idle, while others burst with activity. Randomization ensures the simulation covers a diverse range of temporal patterns, customer profiles,

and stress scenarios, making observed results more representative and statistically robust. This design eliminates artificial periodicity or bias that could arise from fixed transaction schedules, ensuring that both the fuzzy membership statistics and the system's anomaly detection capabilities adapt to realistic, unpredictable workloads.

Each transaction amount—whether a deposit or withdrawal—is drawn from a distribution that evolves over time for each customer, enabling the fuzzy logic model to adapt dynamically. The triangular membership functions are recalibrated using rolling statistics, specifically updating the minimum, maximum, and mean values only for those customers whose transactions are classified as "Highly Suspicious" or "Borderline" as new data is received. This approach ensures that transactions which are unrelated to the flagged malicious activity can continue to be processed during the affected account's recovery period. Unlike the traditional system, which halts all activities until the recovery is complete, the fuzzy system allows normal operations to proceed for unaffected accounts, minimizing unnecessary disruption across the broader system.

4.2 Metric of Interest: Transactions Possessed During Recovery

The central experimental metric is the number of transactions that can be processed during the recovery window following an anomaly, before a final correction is enforced. This directly measures system resilience: higher numbers indicate that operations continue despite suspicious/malicious events, while zero reflects traditional "freeze-on-alert" approaches.

- **Fuzzy System:** Uses adaptive thresholds (randomized within ranges reflecting the category and tuned in real-time) for how many subsequent transactions are processed before full log-based recovery occurs
- **Traditional System:** Always imposes an immediate block (threshold = 0)

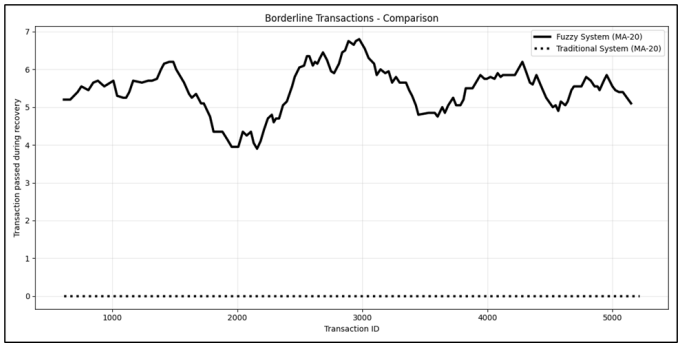
The comparative analysis of "Highly Suspicious" and "Borderline" transaction categories evaluates the effectiveness of the proposed Fuzzy System against a classical traditional system. In the traditional system, global halts are triggered, meaning that once the IDS flags a transaction as malicious, system operations are suspended entirely and a comprehensive log review is initiated to support recovery procedures. This contrasts with the adaptive approach of the Fuzzy System, which aims to maintain operational continuity while managing potential threats.

Since transaction volumes during recovery periods are highly unpredictable and often behave in a random manner, raw counts alone fail to reflect meaningful patterns due to excessive volatility and noise. To address this, the analysis focuses exclusively on the 20-point moving average (MA-20) of permitted transactions during recovery intervals under both the adaptive fuzzy system and the traditional global halt approach.

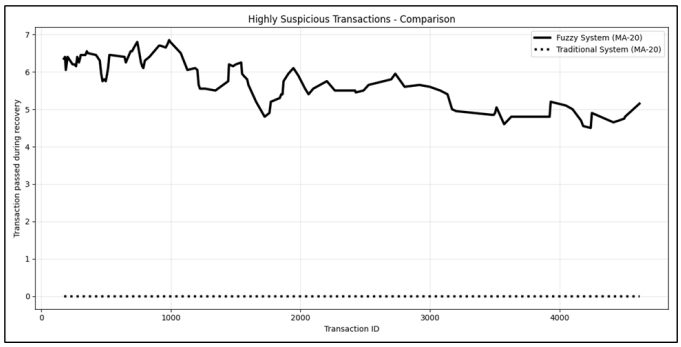
A 20-point moving average (MA-20) is widely adopted in statistical and signal processing literature for trend elucidation. It smooths transient anomalies and random spikes, permitting long-term patterns and realistic averages in system throughput to be visualized and compared. This window size provides balance—suppressing high-frequency noise without losing the ability to track shifts in mean recovery thresholds as attack rates or operational conditions change—and reduces the effect of momentary, irregular spikes.

4.3 Experimental Results

4.3.1 Data Collection and Processing. For each incident, the number of transactions permitted during ongoing recovery was tracked for both the adaptive fuzzy system and the traditional global halt system. For clarity and methodological rigor, only the 20-point moving average (MA-20) values are reported, excluding raw data, which was found to be excessively volatile and unsuitable for meaningful pattern identification.



(a) Border line



(b) Highly Suspicious

Figure 4: Transactions – Recovery Throughput Comparison

4.3.2 Performance Analysis. The visualization in Figures 4a and **??** demonstrate a significant reduction in operational paralysis with the adaptive fuzzy design: the traditional system enforces a complete halt in activities following detection of malicious events, while the fuzzy system maintains substantial transaction throughput even under adverse conditions. The persistence of nonzero values in the moving average curve, observed consistently across thousands of incidents, highlights the robustness of the proposed fuzzy system. This resilience is evidenced by two key features:

(1) It rigorously safeguards the accuracy and integrity of account balances, preventing malicious or erroneous transactions from undermining the correctness of database records.

(2) It supports operational continuity by dynamically adapting its behavior to evolving threat landscapes, ensuring uninterrupted transaction processing even under adverse conditions.

In summary, the experimental results provide strong quantitative support for the proposed fuzzy approach. The marked increase in permissible throughput over traditional methods is achieved without sacrificing post-event data integrity and while operating under real-time constraints, demonstrating that fuzzy recovery is not only theoretically sound but also operationally effective. The empirical findings confirm that the fuzzy approach dramatically enhances throughput and flexibility compared to legacy models, especially critical in high-impact, fast-moving infrastructure domains.

5 Conclusion

This work introduced a fuzzy recovery framework leveraging dual-processor architecture and triangular membership functions to achieve rapid post-attack restoration while maintaining operational continuity in transactional systems. Simulations across 100 accounts demonstrated that our approach sustains substantial throughput during recovery periods where traditional systems enforce complete halts, effectively balancing speed, resilience, and eventual data integrity. While our fuzzy recovery approach demonstrates significant benefits in maintaining throughput and resilience, its effectiveness is currently shaped by task-specific characteristics and is unsuitable for safety-critical domains like healthcare or aerospace where approximate values could endanger lives and exact precision is mandatory.. Because the framework profiles each customer's or item's historical behavior individually—making anomaly detection and recovery decisions based solely on their past data—it is especially well suited for transactional domains where data items (such as accounts in banking or products in supermarkets) are independent and do not require cross-correlation with other records. This "data-centric" strategy works well for environments like banking, retail, or inventory management systems, where each customer or item can be evaluated in isolation.

However, many real-world systems—such as supply chain networks, fraud detection, or interconnected service platforms—involve complex interdependencies among transactions and entities. Extending this framework to handle shared, interconnected, or multi-entity transactions (i.e., moving from data-item-centric to truly transaction-centric or relational analysis) remains an important direction for future work. Generalizing the detection and recovery logic beyond per-customer histories would increase the applicability and robustness of the approach in broader domains and more integrated systems.

References

[1] Wasif Afzal, Richard Torkar, and Robert Feldt. 2009. A systematic review of search-based testing for non-functional system properties. *Information and Software Technology* 51, 6 (June 2009), 957–976. doi:10.1016/j.infsof.2008.12.005

[2] Mariha Siddika Ahmad and Brajendra Panda. 2024. Damage Assessment in Fog Computing Systems: A Blind Write Lineage Approach. In *Proceedings of the IEEE 24th International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW '24)*. IEEE Computer Society, Washington, DC, USA, 50–55. doi:10.1109/CCGridW63211.2024.00012

[3] Mariha Siddika Ahmad and Brajendra Panda. 2024. Validating Damage Assessment: A Simulation-Based Analysis of Blind Write Lineage in Fog Computing. In *Proceedings of the Eighteenth International Conference on Emerging Security Information, Systems and Technologies (SECURWARE '24)*. IARIA, Nice, France, 28–35. https://www.thinkmind.org/library/SECURWARE/SECURWARE_2024/securware_2024_2_10_30009.html

[4] Mariha Siddika Ahmad and Brajendra Nath Panda. 2024. ClusteredLog: Optimizing Log Structures for Efficient Data Recovery and Integrity Management in Database Systems. *Electronics* 13, 23 (Nov. 2024), 4723. doi:10.3390/electronics13234723

[5] Nedhal A. Al-Sayid and Dana Aldlaeen. 2013. Database security threats: A survey study. In *Proceedings of the 5th International Conference on Computer Science and Information Technology (CSIT '13)*. IEEE Computer Society, Washington, DC, USA, 60–64. doi:10.1109/CSIT.2013.6588759

[6] Mansour Alali, Ahmad Almogren, Mohammad Mehedi Hassan, Iehab A. L. Rassan, and Md Zakirul Alam Bhuiyan. 2018. Improving risk assessment model of cyber security using fuzzy logic inference system. *Computers & Security* 74 (April 2018), 323–339. doi:10.1016/j.cose.2017.09.011

[7] Paul Ammann, Sushil Jajodia, Catherine D McCollum, and Barbara T Blaustein. 1997. Surviving information warfare attacks on databases. In *Proceedings. 1997 IEEE Symposium on Security and Privacy (Cat. No. 97CB36097)*. IEEE, 164–174.

[8] Hoyong Choi and Sangjin Lee. 2023. Forensic analysis of SQL server transaction log in unallocated area of file system. *Forensic Science International: Digital Investigation* 46 (2023), 301605. doi:10.1016/j.fsidi.2023.301605

[9] Hoyong Choi, Sangjin Lee, and Doowon Jeong. 2021. Forensic Recovery of SQL Server Database: Practical Approach. *IEEE Access* 9 (2021), 14564–14575. doi:10.1109/ACCESS.2021.3052505

[10] Shilin He, Jieming Zhu, Pinjia He, and Michael R. Lyu. 2016. Experience Report: System Log Analysis for Anomaly Detection. In *2016 IEEE 27th International Symposium on Software Reliability Engineering (ISSRE)*. 207–218. doi:10.1109/ISSRE.2016.21

[11] Mamoona Humayun, NZ Jhanjhi, Maram Fahhad Almufareh, and Muhammad Ibrahim Khalil. 2022. Security Threat and Vulnerability Assessment and Measurement in Secure Software Development. *Computers, Materials & Continua* 71, 3 (2022), 5039–5059. doi:10.32604/cmc.2022.019289

[12] Mamoona Humayun, Noor Zaman Jhanjhi, and Abdullah Almotilag. 2022. Real-Time Security Health and Privacy Monitoring for Saudi Highways Using Cutting-Edge Technologies. *Applied Sciences* 12, 4 (Feb. 2022), 2177. doi:10.3390/app12042177

[13] Marina Soledad Iantorno and Khalil Beladda. 2025. Fuzzy Logic for Cybersecurity: Intrusion Detection and Privacy Preservation with Synthetic Data. In *Proceedings of the 17th International Conference on Agents and Artificial Intelligence (ICAART '25)*. INSTICC, SciTePress, Setúbal, Portugal, 376–382. doi:10.5220/0013137300003890

[14] Asif Iqbal, Siffat Ullah Khan, Mahmood Niazi, Mamoona Humayun, Najm Us Sama, Arif Ali Khan, and Aakash Ahmad. 2024. Advancing database security: a comprehensive systematic mapping study of potential challenges. *Wireless Networks* 30, 7 (Oct. 2024), 6399–6426. doi:10.1007/s11276-023-03436-z

[15] Sanaa Kaddoura, Ramzi A. Haraty, Karam Al Kontar, and Omar Alfandi. 2021. A Parallelized Database Damage Assessment Approach after Cyberattack for Healthcare Systems. *Future Internet* 13, 4 (April 2021), 90. doi:10.3390/fi13040090

[16] A. Kupchyn, V. Komarov, I. Borokhvostov, M. Bilokur, A. Kuprinenko, Y. Mishchenko, V. Bohdanovych, and O. Kononov. 2022. Determining the Accuracy for Fuzzy Logic Technology Foresight Model. *Cybernetics and Systems Analysis* 58, 3 (May 2022), 382–391. doi:10.1007/s10559-022-00470-1

[17] Peng Liu, Paul Ammann, and Sushil Jajodia. 2000. *Rewriting Histories: Recovering from Malicious Transactions.* Springer US, Boston, MA, 7–40. doi:10.1007/978-1-4615-4461-6_2

[18] David R. Matos, Ibéria Medeiros, Miguel L. Pardal, and Miguel Correia. 2024. *Intrusion Recovery Systems: A survey.* Technical Report. Instituto Superior Técnico, Universidade de Lisboa. doi:10.36227/techrxiv.172296768.82082288/v1 Submitted to IEEE Transactions.

[19] Mahfuzul I. Nissan. 2025. *MemTraceDB: Reconstructing MySQL User Activity Using ActiviTimeTrace Algorithm.* arXiv:2509.05891 [cs.CR] https://arxiv.org/abs/2509.05891

[20] Yihao Peng, Biao Ma, Hai Wan, and Xibin Zhao. 2025. *Ancora: Accurate Intrusion Recovery for Web Applications.* arXiv:2510.07806 [cs.CR] https://arxiv.org/abs/2510.07806

[21] Ihor Subach, Vitalii Fesokha, Artem Mykytiuk, Volodymyr Kubrak, and Stanislav Korotayev. 2021. Simulation Model of a Fuzzy Cyber Attack Detection System. In *Proceedings of the XXI International Scientific and Practical Conference Information Technologies and Security (ITS '21)*. CEUR-WS.org, Aachen, Germany, 92–101. https://ceur-ws.org/Vol-3241/paper9.pdf